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Georgios Avdis

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In "Implicit Regularization of Infinitesimally-perturbed Gradient Descent Toward Low-dimensional Solutions", Ma et. al. give a nice strategy for studying gradient descent with leave-one-out analysis.

- **Step 1:** Local Geometry: smoothness and strong convexity (uniform in some region).
- **Step 2:** Introduce leave-one-out sequences.
- **Step 3:** Establish "general incoherence" (depends on the problem at hand)
  (a) Show that the true iterates and the leave-one-out iterates are close.
  (b) Use the independence to establish incoherence of the leave-one-out iterates.
  (c) Combine the above.

$y_j = \psi_j H X^\top \phi_j$, $1 \leq j \leq m$
In matrix completion, we have that:

$$\begin{aligned} \psi_j &= e_k \\ \phi_j &= e_l \\ m &= |\Omega| \\ H &= X \end{aligned}$$

while in phase retrieval we have

$$\begin{aligned} \psi_j &= a_j \\ \phi_j &= b_j \end{aligned}$$

where, for example, we might have that $a_j, b_j \sim \mathcal{N}(0,1)$.

When establishing the general incoherence, we need to control

$$\max_j ||\psi_j^\top (H^t - H)||$$
$$\max_j ||\phi_j^\top (X^t - X)||$$

Iterates are "mis-aligned", ensures entrywise convergence.

This could be generalized in the following way:

In the first step, we don't need strong convergence. We could use alternative conditions instead of $\nabla^2 f(x)[V,V]$ (e.g. eigengap for tensors).

Also, $\nabla^2 f(x)[V,V]$ might be rank-deficient, but could be "well-behaved".

In overparameterized matrix sensing we could use a combination of Restricted Strong Convergence and taking advantage of some case specific nullspace behavior.

$y_j = Tr(\phi_j \psi_j^\top H X^\top) =< \phi_j \psi_j^\top, H X^\top >$.

LASSO

$$
\begin{aligned}
y_i &= \beta^\top X_i + \varepsilon_i \\
\varepsilon_i &\sim \mathcal{N}(0, \sigma^2) \\
X_i &\sim \mathcal{N}(0, \Sigma)
\end{aligned}
$$

In LASSO, the objective function is

$$h(\beta) = \frac{1}{2n} \sum_{i=1}^{n} (y_i - X_i^\top \beta)^2 + \lambda ||\beta||_1$$

Notice how this is similar to the problem of symmetric matrix completion where the objective function was

$$\frac{1}{n} \sum_{(i,j)\in\Omega} (M_{ij} - Z_{ij})^2 + \lambda ||Z||_*$$

The KKT conditions are equivalent to the first order necessary conditions for local (global) minima and

$$h^{'}(\beta) = \frac{-X^\top}{n}(Y - X\beta) + \lambda \cdot \operatorname{sgn}(\beta)$$

so the solution $\hat{\beta}_L = \arg\min h(\beta)$ must satisfy that

$$
\begin{aligned}
\frac{X^\top}{n}(X\beta + \varepsilon - X\hat{\beta}_L) &= \lambda \cdot \operatorname{sgn}(\hat{\beta}_L) \\
\frac{X^\top X}{n}(\beta - \hat{\beta}_L) + \frac{X^\top \varepsilon}{n} &= \lambda \cdot \operatorname{sgn}(\hat{\beta}_L) \\
\beta - \hat{\beta}_L + \left(\frac{X^\top X}{n}\right)^+ \frac{X^\top \varepsilon}{n} &= \lambda \left(\frac{X^\top X}{n}\right)^+ \operatorname{sgn}(\hat{\beta}_L) \\
\beta + \left(\frac{X^\top X}{n}\right)^+ \frac{X^\top \varepsilon}{n} &= \hat{\beta}_L + \lambda \left(\frac{X^\top X}{n}\right)^+ \operatorname{sgn}(\hat{\beta}_L) \\
\beta + \left(\frac{X^\top X}{n}\right)^+ \frac{X^\top \varepsilon}{n} &= \hat{\beta}_L + \left(\frac{X^\top X}{n}\right)^+ \frac{X^\top}{n}(Y - X\hat{\beta}_L) \\
&:= \hat{\beta}_D
\end{aligned}
$$

where $\hat{\beta}_D$ is the debiased LASSO estimator for β.

We have that:

$$\begin{aligned}\hat{\beta}_D - \beta &= \left(\frac{X^\top X}{n}\right)^+ \frac{X^\top \varepsilon}{n} \\ &= \Sigma^{-1}\frac{X^\top \varepsilon}{n} + \left(\left(\frac{X^\top X}{n}\right)^+ - \Sigma^{-1}\right)\frac{X^\top \varepsilon}{n}\end{aligned}$$

but because $X \sim \mathcal{N}(0, \Sigma)$, the second term can be proven to be negligible and every row has the property that

$$e_j^\top(\hat{\beta}_D - \beta) \approx \mathcal{N}(0, \frac{\sigma^2}{\Sigma_{jj}})$$

as $n, p \to \infty$ and provided that λ is chosen accordingly and $s = o(\frac{\sqrt{n}}{\log^2(p)})$ (the amount of bias introduces is "small").

The second term is bounded as

$$||e_j^\top\left(\left(\frac{X^\top X}{n}\right)^+ - \Sigma^{-1}\right)\frac{X^\top \varepsilon}{n}|| \leq ||\left(\frac{X^\top X}{n}\right)^+ - \Sigma^{-1}||_\infty ||\frac{X^\top \varepsilon}{n}||_1$$

and then by using some sparsity condition on Σ.

In the matrix denoising case, we observe $\boldsymbol{M} + \boldsymbol{N}$ where $\boldsymbol{M}$ is rank-r amd $\boldsymbol{N}$ is noise.

We then want to minimize the function

$$h(\boldsymbol{Z}) = ||\boldsymbol{M} + \boldsymbol{N} - \boldsymbol{Z}||_F^2$$

across matrices $\boldsymbol{Z}$ such that $rank(\boldsymbol{Z}) \leq r$, which is equivalent with minimizing

$$h(\boldsymbol{X}) = ||\boldsymbol{M} + \boldsymbol{N} - \boldsymbol{X}\boldsymbol{X}^\top||_F^2$$

across matrices $\boldsymbol{X}$ such that $\boldsymbol{X} \in \mathbb{R}^{n \times r}$.

But, if r is unknown, we can let

$$h(\boldsymbol{Z}) = ||\boldsymbol{M} + \boldsymbol{N} - \boldsymbol{Z}||_F^2 + \lambda \sum_{i=1}^{n} \mathbb{1}_{\{\sigma_i(\boldsymbol{Z}) > 0\}}$$

or

$$h(\boldsymbol{Z}) = ||\boldsymbol{M} + \boldsymbol{N} - \boldsymbol{Z}||_F^2 + \lambda ||\boldsymbol{Z}||_*$$

In that case, the optimal $\boldsymbol{Z}$ is the soft threshold:

$$\boldsymbol{Z}_{opt} = sh_\lambda(\boldsymbol{M} + \boldsymbol{N}) = \hat{U}(\hat{\Sigma} - \lambda I) + (\hat{V})^\top$$

where

$$\boldsymbol{M} + \boldsymbol{N} = \hat{U}\hat{\Sigma}(\hat{V})^\top$$

This estimator is biased, but we can try and debiase it.

The singular values of $\boldsymbol{Z}_{opt}$ are
$\{\sigma_1(M+N), \sigma_2(M+N), ..., 0\}$
By adding λ we get: $\hat{U}\tilde{\Sigma}(\hat{V})^\top$ and $\tilde{\Sigma}$ only has singular values that are greater than λ.

Now, if $\sigma_{r+1}(M+N) < \lambda < \sigma_r(M+N)$, then the estimator will have the correct rank.

In the non-convex scenario, the objective function becomes:

$$h^{nc}(\boldsymbol{X}) = \frac{1}{4}||\mathcal{P}_\Omega(\boldsymbol{M}+\boldsymbol{N}-\boldsymbol{X}\boldsymbol{X}^\top)||_F^2 + \lambda||\boldsymbol{X}||_F^2$$

with the convex one being:

$$g^c(\boldsymbol{Z}) = \frac{1}{2}||\mathcal{P}_\Omega(\boldsymbol{M}+\boldsymbol{N}-\boldsymbol{Z})||_F^2 + \lambda||\boldsymbol{Z}||_*^2$$

We have shown in previous lectures that the minimizers of the above two functions are the same. For the debiased version we solve:

$$\left(g^c(\boldsymbol{Z})\right)' = -\mathcal{P}_\Omega(\boldsymbol{M}+\boldsymbol{N}-\boldsymbol{Z}) + \lambda\frac{\partial||\boldsymbol{Z}||_*}{\partial \boldsymbol{Z}} = 0 \implies$$
$$\mathcal{P}_\Omega(\boldsymbol{M}+\boldsymbol{N}-\boldsymbol{Z}) = \lambda\frac{\partial||\boldsymbol{Z}||_*}{\partial \boldsymbol{Z}}$$

We then can write:

$$(\mathcal{P}_\Omega - p\mathcal{I})(M+N-Z_{opt}) + p(M+N-Z_{opt}) = \lambda V$$

or

$$\boldsymbol{M}+\boldsymbol{N} = \boldsymbol{Z}_{opt} + (\mathcal{I} - \frac{\mathcal{P}_\Omega}{p})(M+N-Z_{opt}) + \frac{\lambda V}{p}$$

which is very similar to the debiased LASSO case. Here, the debiased estimator can be written as:

$$\begin{aligned}\boldsymbol{Z}_D &= -alP_r(\boldsymbol{Z}_{opt} + \frac{1}{p}\mathcal{P}_\Omega(\boldsymbol{M}+\boldsymbol{N}-\boldsymbol{Z})) \\ &= \mathcal{P}_r(\frac{1}{p}\mathcal{P}_\Omega(\boldsymbol{M}) + (\mathcal{I}-\frac{1}{p})\boldsymbol{Z} + \frac{1}{p}\mathcal{P}_\Omega(\boldsymbol{N}))\end{aligned}$$

and argue that the first term behaves similarly to $\boldsymbol{M}$, the second one is small (similar analysis as in the debiased LASSO estimator) and that the third one is just noise.

Question: Can we say that $\boldsymbol{Z}_D$ is also optimal for the non-convex case?

Probably yes, but the non-convex estimator might already be asymptotically unbiased by default.